

Exploring the Ethical Dimensions of Engineering Processes for Solar Energy Production

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Abstract—Global manufacturing of photovoltaic (PV) cells has increased substantially over the last two decades. From the installation of 1.5 GW to over 580 GW of PV capacity, solar power has become one of the most prominent renewable energy sources in the world [1]. The cost to manufacture PV systems has fallen by as much as 95% since 2010, contributing to its surge in production and its competitiveness in the global economy [2].

Despite the promising figures and alluring potential of investing in solar energy to meet contemporary green energy demands, a series of ethical flaws challenges its long-term value as a sustainable energy source. Numerous allegations of forced labor and the utilization of fossil fuels as a cheaper alternative to cleaner energy in the manufacturing process only scratch the surface of the complex ethical landscape of solar energy production [3].

This paper explores the ethical dimensions of harnessing solar power for green energy initiatives and how they align with the pillars of ethics in science and engineering research and practice. The analysis considers a blend of modern case studies and characterizes their findings to identify key alternatives to address the ethical pitfalls in PV manufacturing such as a lack of transparency in the origins of raw materials, supply chain issues, human rights violations, and accountability for policy regulations. The discussion uses these alternatives to formulate a timeline of actions stakeholders can take to balance the incentive to manufacture PV systems at scale with the obligation to uphold ethical standards in the production lifecycle.

Index Terms—Green Energy; Ethics; Renewable Energy; Solar Panel Manufacturing

I. INTRODUCTION

The increasing concern to combat climate change has been fueled by an international push for green energy spanning across the whole spectrum of users from commercial farms to personal use. One of the most widely utilized renewable energy sources comes from harnessing the energy from the sun by way of electricity in a solar

cell. “Solar energy is the fastest growing renewable energy source” [4] and this is due to the evolution of the first photovoltaic (PV) cell approximately one-hundred and fifty years ago.

Solar panels create energy through a reaction of visible light by the radiation interacting with the semiconducting coating; a process that is identified as the “photovoltaic effect” [4]. This effect is created through an array of PV cells which allow for the photons from the sunlight to be passed through and absorbed into the semiconductor material and initiate the dislodging of electrons from the semiconductor material’s atoms. The flow of electrons then creates a voltage potential that when connected to an electrical circuit allows for electricity to flow. While PV cells can be applied to several systems ranging in amount of electricity necessary, the current evolutions surrounding this technology looks at increasing the efficiency while ensuring the current market can keep up with the world’s solar demands.

The companies currently at the top of the PV field have a litany of ethical concerns surrounding undercutting Western prices by way of cheap labor, cheap manufacturing [5], and supply chain issues. In fact the “production of almost half of the world’s supply of polysilicon, a critical component of solar panels, concentrated in Xinjiang, where Beijing-backed human rights abuses were rife, fed opposition to reliance on Chinese solar technology” [6]. As the demand increases for solar energy and the PV cells required, and the forced labor concerns are illuminated, countries have started adopting policies and legislation to address the conditions to correct the problems on an individual basis. As china maintains its hold on the PV cell market, they can continue to set their own rules when it comes to how their workers are treated at their manufacturing plants.

Until the newer companies can produce PV cells at the same efficiency and comparative cost as the Chinese plants, solutions to these ethical issues appear far-off.

“Engineers at all times should strive to serve the public interest and are encouraged to adhere to the principles of sustainable development in order to protect the environment for future generations” [7]. Sustainable development also includes meeting the human needs on a personal basis and therefore manufacturing plants should not be prioritizing their profits over the wellbeing and safety of its workers. The manufacturing of PV cells provides a prime example to explore the relationship between a currently coveted engineering technology and the ethical dilemmas inhibiting meeting the market demand.

II. ETHICAL CHALLENGES IN PV MANUFACTURING

“A photovoltaic (PV) cell, commonly called a solar cell, is a nonmechanical device that converts sunlight directly into electricity ... A PV cell is made of semiconductor material. When photons strike a PV cell, they may reflect off the cell, pass through the cell, or be absorbed by the semiconductor material. Only the absorbed photons provide energy to generate electricity ... The PV cell is the basic building block of a PV system.” Since the inception of PV cells, solar energy has emerged as the fastest-growing energy source, expanding at a remarkable rate of 24% annually. Despite this growth, the industry is grappling with a range of ethical challenges, including supply chain issues, forced labor, and the management of end-of-life PV cell disposal.

Now, PV cells don't last forever as the life cycle of PV cells lasts around 30-35 years. Once the PV cell reaches the end of its life, there are multiple ways that it will be decommissioned: Repair and Reuse, Refurbish/Remanufacture, Recycling, or in Landfills. The process for a PV cell if it gets repaired and reused is that those service providers will repair and distribute the panels to other projects. However, many challenges arise from here as well, as the panels will require more inspection, repair, testing, and, sometimes, recertification. Next, is that cells can get refurbished or remanufactured. Manufacturers can reclaim the panels to further extend the lifetime of the cells and/or their components. However, this path also runs into many of the same economic and regulatory problems as repair and reuse. The most “wasteful”, unsustainable, and easiest option, is through e-waste. “The World Economic Forum estimates \$62.5 billion worth of valuable materials are locked up in e-waste each year.” This method traps these materials underground and out of reach. However, most of the e-waste ends up in many poorer nations to be “taken care of.” Around 50 million metric tons of e-waste is generated globally each year. Another option that is taken is through recycling where once the electronics and frame are removed, the panel is cut into two pieces, its silicon and

CdTe Thin Film bases. The rest of the process is explained in Figure 1.

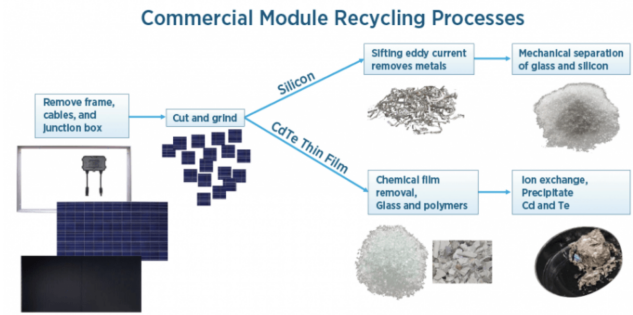


Fig. 1: Recycling process for commercial solar panels

We have the ethical issues of improper end-of-life care for many of our electronics, including PV cells, but the most severe ethical issues exist in a different way. Currently, some of the largest systems exist in the United States, Italy, Germany, and, most importantly, China. Within the last few years, many human rights allegations have been made in China, specifically in the Xinjiang region. It starts with domination that China currently has over the manufacturing components of solar panels, including polysilicon, wafers, cells, and modules (expanded upon in Figure 2).

China's Solar Dominance

Beijing holds a strong control over all the manufacturing components of solar panels

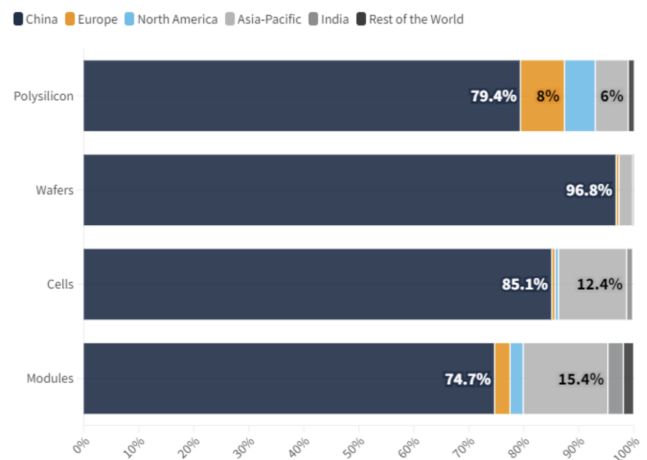


Fig. 2: Share of solar manufacturing components by country

Next, according to a UN report, the Chinese government has committed “serious human rights violations” against the Uyghur population. “In 2021 academics at Sheffield Hallam University said the the biggest polysilicon producers in the region [Xinjiang] reported their participation in ‘labor transfer’ programmes administered by ‘in an environment of unprecedented coercion.’” However, the Chinese government disputes the allegations of forced labor, arguing that employment is voluntary. Since many nations, including the US, rely on the supply chain in China, this could

jeopardize each nation's own supply chains. Though the US has made sanctions on the Xinjiang companies, they are continuing their work, skirting trade rules, and making moves to expand.

The stakeholders involved in this complex system encompass PV manufacturers, governments, and consumers. PV manufacturers are tasked with sourcing materials and labor ethically while maintaining cost efficiency. Governments play a pivotal role in balancing environmental imperatives with human rights considerations, exemplified by initiatives like Japan's solar panel mandate aimed at achieving carbon neutrality. The US Uyghur Forced Labor Prevention Act demonstrates governmental efforts to address ethical concerns within the PV supply chain, emphasizing the importance of accountability and transparency.

Consumers also wield influence as stakeholders, demanding products sourced sustainably and ethically. The ethical dimensions of PV cell manufacturing and disposal resonate with environmentally conscious consumers, driving demand for greater transparency and accountability across the industry.

One of the largest effects was in the Tokyo Solar Panel Mandate which requires home builders and developers to install solar panels on new buildings and houses starting from the spring of 2025. This is moving towards the goal Japan has to reach carbon neutrality. However, to reduce costs, these developers may look to China, as will most of the world. Tokyo Governor Koike needs to make a plan for ensuring the solar panels are not tainted by the human rights violations in China. Still, Koike's responses have not been so reassuring.

In conclusion, addressing the multifaceted ethical challenges within the PV industry demands concerted global action and collaboration. From responsible end-of-life management to ethical supply chain practices, stakeholders must prioritize sustainability, transparency, and human rights to ensure the continued growth of solar energy as a clean and ethical energy source.

III. ANALYSIS THROUGH THE LENS OF THE NSPE CODE OF ETHICS FOR ENGINEERS

Engineers face many ethical issues as a result of China's dominant position in the solar panel trade, especially in the areas of safety, health, welfare, ethical design, sustainable development, privacy protection, and timely disclosure of potential concerns. The concentration of solar manufacturing and polysilicon capacity in China, together with allegations of forced labor in the sector, raises questions about environmental justice, human rights abuses, and ethical sourcing.

As Engineers we use the NSPE code of ethics as a base to form our actions around. We use the clauses to define what's morally just.

I. To uphold the highest standards of integrity, responsible behavior, and ethical conduct in professional activities. [7]

- *To hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment; [7]*
- *To avoid unlawful conduct in professional activities, and to reject bribery in all its forms. [7]*

II. To treat all persons fairly and with respect, to not engage in harassment or discrimination, and to avoid injuring others. [7]

- *To treat all persons fairly and with respect, and to not engage in discrimination based on characteristics such as race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression; [7]*
- *To avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors or any other verbal or physical abuses. [7]*

Concerns regarding equitable treatment and respect for all people are raised by reports of forced detention, labor, and human rights violations, notably pertaining to Uyghur and other minority groups in China's solar business. A system that participates in discriminatory practices and human rights violations may be indirectly supported or beneficial to engineers and businesses in this field. The ethical principle of treating all persons fairly and with respect, without engaging in discrimination based on various characteristics, is compromised in the context of China's dominance in the polysilicon and solar industry, especially concerning the reported human rights abuses against Uyghurs and other minority groups. Reports of forced detention and labor of Uyghurs and other minority groups in China raise concerns about inadvertently supporting or benefiting from a system that engages in discriminatory practices through forced labor. The solar industry's engineers and stakeholders need to confront breaches of human rights because forced labor is in direct opposition to the values of ethical sourcing, safety, health, and welfare. With China leading the way in the polysilicon and solar industries, there is a risk that the engineer's duty to treat everyone equally and with respect and to refrain from damaging anyone else's property, reputation, or employment through deceptive or malicious behavior will be compromised, especially in light of documented violations of human rights.

Promoting Human Rights and Ethical Sourcing Engineers have an ethical duty to uphold fairness and refrain from prejudice, which includes supporting social justice and human rights. Upholding ethical standards in the solar business of China necessitates strong initiatives, transparency, and international cooperation in resolving documented human rights abuses.

Supply chain accountability and transparency are necessary to guarantee equitable treatment guarantee that everyone in the supply chain—including the laborers who make solar panels and polysilicon—is treated fairly and with respect. Effectively upholding these ethical norms can be challenging, though, due to the complexity of the global supply chain and the difficulties in tracking down the sources of commodities and labor conditions. Traceability are critical for engineers and businesses to identify and reduce hazards including forced labor, environmental damage, and unethical business practices.

Businesses in the solar sector, particularly those with strong connections to China, are obliged to adhere to CSR(Corporate Social Responsibility) principles, which include treating employees fairly, respecting human rights, and preventing harm to stakeholders and communities. However, compromises in CSR policies may result from conflicting interests and financial constraints.

The global market's ability to diversify is hampered by China's predominant role in solar manufacturing. This industry's resilience, innovation, and competition may be hampered by this concentration of manufacturing capacity. It is recommended that engineers and other relevant parties investigate means of expanding their production base beyond China in order to foster a more equitable and sustainable marketplace. The market climate created by China's dominant control over polysilicon and solar production may lead to market dominance and dependence, perhaps pressuring enterprises and engineers to resort to unethical activities in order to gain contracts or preserve competitiveness. This can entail paying no attention to abuses of human rights or engaging in bribery to obtain an unfair advantage in the marketplace. Due to China's hegemony in the solar and polysilicon industries, there is a possibility that businesses and individuals will be under pressure to put profits ahead of morality. This pressure may force people to make concessions on justice, decency, and protecting the interests of others.

Decision-making processes may be influenced by the geopolitical environment and the financial incentives associated with China's hegemony in the solar manufacturing industry. Governments, investors, or trade associations may put pressure on engineers and businesses to put cost-effectiveness and market access ahead of moral considerations like respect for human rights and anti-corruption initiatives.

It's possible that the solar industry's current tracing procedures and monitoring systems are insufficient to identify and deal with illegal activity or bribes. The efforts of engineers and ethical practitioners may be undermined by this ineffective oversight, which leaves room for unethical activity to continue.

In the solar supply chain, advocates such as advocacy groups, legislators, business executives, and engineers are essential in promoting respect, equity, and moral behavior.

To address human rights problems, promote transparency, and sustain ethical standards in the industry, cooperation, communication, and coordinated actions are required.

The environmental impact of solar energy has decreased due to China's dominance in the industry, but worries about the long-term effects are still there. Engineers must limit environmental damage throughout the solar panel lifecycle, from raw material extraction to end-of-life disposal, in accordance with sustainable development standards. This entails dealing with problems including waste management, resource depletion, and pollution.

Ultimately, the moral dilemmas raised by China's involvement in the solar panel industry necessitate a multifaceted strategy that places an emphasis on environmental sustainability, market diversification, transparency, and human rights. In order to promote moral behavior, guarantee supply chain transparency, and spur innovation in the direction of a more just and sustainable solar industry, engineers are essential. Given these difficulties, engineers must navigate a challenging ethical environment where it is critical to strike a balance between their professional obligations and the demands of the market and geopolitical factors. Reforming the solar supply chain, advancing transparency, fortifying oversight mechanisms, and advocating for moral norms necessitate cooperation between governments, business executives, civil society organizations, and international stakeholders. A multi-stakeholder approach, accountability, openness, and a dedication to advancing social justice and human rights in the solar supply chain are necessary to address these ethical issues.

In summary, given the dominant position of China in the polysilicon and solar industries, particularly with regard to documented human rights abuses, the engineer's duty to treat everyone equally, with respect, and to refrain from harming others by deceitful or malevolent activities, may provide difficulties and compromises. It will need a team effort, moral leadership, openness, and a dedication to respecting human rights and moral standards in engineering practice and corporate operations to overcome these obstacles.

IV. STRATEGIES AND APPROACHES FOR ETHICAL PV MANUFACTURING

The ethical landscape of PV manufacturing is flawed, but not beyond repair. Multiple case studies have shown how businesses can adopt plausible solutions to improve the industry's integrity and protect its workers, local communities, and the environment. But these solutions are only substantive if they are actionable and conform to modern ethical frameworks in science and engineering. To develop these solutions, this section explores a pair of case studies and their proposed solutions to describe their practicality and broader applications that can be achievable in the near-term, mid-term, or far-term.

The first case study by Sovacool et al. (2022) [8] probes the “energy injustices” associated with household adoption of PV cells within the Brighton neighborhood of the United Kingdom’s East Sussex County. Their analysis introduced a taxonomy of inequities in household adoption of PV cells for solar energy: demographic, spatial, interspecies, and temporal. These inequities exemplify a subset of the ethical challenges introduced earlier in this report to encapsulate the energy injustices existing in the industry. The authors found high interconnectivity between these inequities, and Figure 3 shows their relationships paired with areas the authors considered to be the most highly connected.

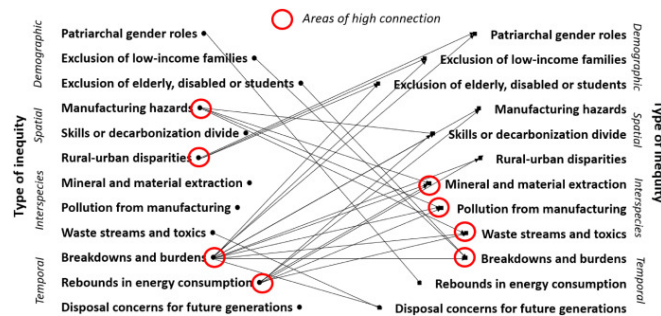


Fig. 3: From Sovacool et al. (2022)

After identifying these inequities at the local scale, the authors discuss a set of solutions to inform the development of future policies to resolve these inequities. For example, they recommend that local governments should call for “improved governance and accountability in supply chains and labor standards” to remedy the ethical gaps reaching into all four inequities they unveiled at the local scale. Businesses can achieve this goal in the far-term by working with local governments to introduce legislation focused on stricter oversight in the PV supply chain.

The authors also introduced a method to resolve demographic inequities by recommending “specific amendments to [the UK’s] Modern Slavery Act of 2015 to target all low-carbon technologies ...to ensure the transition to a low-carbon economy ...is not directly connected to poor, illegal, or inhumane labor practices within solar supply chains.” Businesses can adopt this solution in the mid to far-term due to the nature of the time it takes for these amendments to pass and take effect. To counter spatial inequities, the authors propose “policy incentives that lower the cost of rural adoption along with targeted interventions that prioritize areas high in deprivation.” This is another solution that businesses and governments can adopt in the mid to far-term as the development of policy incentives with substantial economic impact requires a holistic approach to work successfully.

Finally, the authors introduce the concept of “shared ownership business models, including cooperatives or council-led schemes that offer solar deployment and maintenance to homes unable to pay” to provide another solution

for demographic inequity. Businesses can begin working today on adopting a collaborative business model with peers in their industry, making this solution highly achievable in the short-term.

The second case study by Heffron et al. (2021) [9] looks at solar energy through the lens of existing energy laws aimed at achieving goals set forth in the 2015 Paris Climate Agreement. A recurring theme in their analysis is the growing need for laws protecting vulnerable populations from the exploitation of resources and supply chain laws in PV manufacturing. They studied 72 countries and their introduction of energy law accounting for the ethics and manufacturing requirements of harnessing solar energy. As an example, they argue that citizens should expect to be supplied with clean energy in a just way if it is considered economic and affordable, but that these citizens do not have a sufficient understanding of why they should have this expectation due to literature gaps in modern energy policy.

Through identifying these gaps, the authors share seven principles of energy law comprising their ethical foundations. They use these principles with the current economic and legal implications of energy laws in the 72 countries they studied to propose a solution that “public decision-making ...guided by different forms of energy justice – such as distributive, procedural, recognition, restorative, and cosmopolitanism – can increase certainty and give investors confidence” in choosing solar energy as a valid energy source for future generations. This highlights the incentive for companies to adopt more ethical approaches to PV manufacturing for solar energy initiatives by suggesting that justice-driven decisions in energy policy development will produce a net positive economic benefit as more countries pursue these initiatives. This is another example of a short-term solution that businesses can adopt to improve the ethical foundations of PV manufacturing to make solar energy more valuable in the future.

To summarize this section, businesses have multiple alternatives at their disposal when striving to improve their ethical posture in PV manufacturing. Figure 4 highlights the alternatives identified from the two case studies, and visualizes an assessment of how achievable these alternatives from a temporal perspective. It is important to note that countless other alternatives exist beyond the two case studies discussed in this section; academic and professional analyses continue to evolve, offering stakeholders contemporary solution sets to consider in their decision-making processes.

V. CONCLUSION

To conclude this study, it is important to acknowledge that the PV manufacturing industry in its current form is ethically flawed. Supply chains carry the weight of documented human rights abuses, labor exploitation, environmental risks, and the looming threat of the displacement of indigenous communities for the extraction

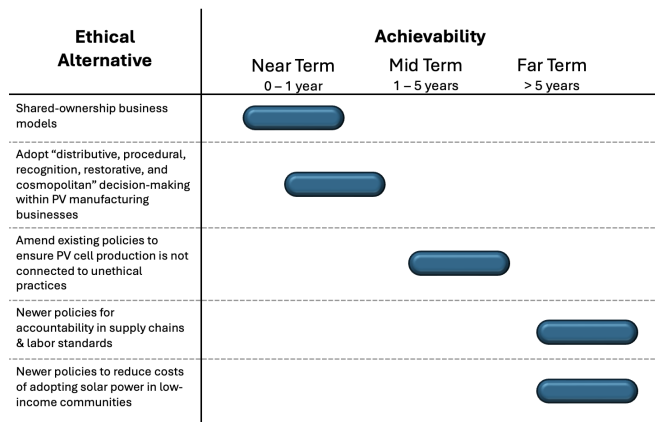


Fig. 4: Assessment of Ethical Alternative Achievability

of raw materials. The source of these ethical drawbacks stems from a combination of inadequate business oversight, muddled supply chains spanning across several countries with diverse labor and environmental regulations, corporate tendencies to prioritize profit at the expense of fair and ethical working conditions, and countless other factors within the social, economic, and political spectrums of the global initiative to address climate change.

The lackluster ethical landscape of PV manufacturing, however, is not irreparable. The goal of this analysis was to first identify specific ethical problems associated with the PV manufacturing industry and address them through contemporary ethical frameworks in science and engineering. It then used these examples to tie with case studies focusing on a subset of these issues and observe their findings to present ethically considerate alternatives to provide to stakeholders. The analysis comments on whether these solutions are achievable in the near-term, mid-term, or far-term to guide the reader toward solutions that are most actionable, from their perspective, to improve the ethical components of the industry.

There are several solutions available to address the ethical issues unveiled earlier in the study. The near-term solutions are to adopt shared-ownership business models as proposed by Sovacool et al. (2022) [8] to synergize PV-focused businesses and the consumer, and to implement Heffron et al.'s [9] justice-driven decision-making strategies at a broader scale in the industry. A mid-term solution includes amending existing energy policy where gaps may exist. Implementing new policies, such as those for more accountability in supply chains and to reduce costs for PV installation in low-income communities, are assessed to be far-term solutions due to the nature of energy policy.

These solutions outline what stakeholders can do today, what they can do in the next five years, and what they can do in the next decade to remedy the ethical problems identified in this analysis. A broader takeaway is that ethical challenges are not constrained to PV manufactur-

development.

ing: electric vehicles (EVs), wind power, hydropower, and other green industries could benefit from this analysis as the proposed solutions may extend to a wider audience that could find utility for them in various manufacturing processes. Future work could involve a temporal study of how businesses implement these solutions, and whether they are successful in the achievability time frames that were specified.

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